ETH zürich

Literature Review

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Overview of literature review

- Literature review of 13 papers
- Main topics: crack propagation in snow, snow modeling, modeling in more general sense (introduction to different modeling methods used)
- Modeling methods: Material Point Method (MPM) and Discrete Element Method (DEM)

Modeling Methods

- Two main methods used: MPM and DEM
- DEM described by Cundall and Strack, 1979, material of disks & spheres, contact forces and resulting displacements calculated
- MPM where material is discretized in points with Eulerian background mesh to calculate stresses at points and time-dependent equations, for example in Gaume et al., 2018
- MPM is more favorable for slope-scale simulations because of modelling of grain contacts
- Other models for snow such as SNOWPACK (Bartelt & Lehning, 2002) and CROCUS (Brun et al., 1992) model snowpack response to external factors

Bibliography

Bartelt, P., & Lehning, M. (2002). A physical SNOWPACK model for the Swiss avalanche warning: Part I: Numerical model. *Cold Regions Science and Technology*, *35*(3), 123–145. https://doi.org/10.1016/S0165-232X(02)00074-5

Brun, E., David, P., Sudul, M., & Brunot, G. (1992). A numerical model to simulate snow-cover stratigraphy for operational avalanche forecasting. *Journal of Glaciology*, *38*(128), 13–22. https://doi.org/10.3189/S0022143000009552

Cundall, P. A., & Strack, O. D. L. (1979). A discrete numerical model for granular assemblies. *Géotechnique*, *29*(1), 47–65. https://doi.org/10.1680/geot.1979.29.1.47

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